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Paper 1 of 1 — MTH 105 MATRICES AND DETERMINANT (2)_Phoenix

Level 100 • First Semester • Handout

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MATRICES AND DETERMINANT

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MATRICES AND DETERMINANT

Matrices and determinant are mainly used for the solution of linear simultaneous equations. The coefficients of the variables for linear simultaneous equations may be shown in matrix form. The coefficient of x and y in the simultaneous equations

$$x + 2y = 3$$

$$4x - 6y = 6$$

Become $\begin{pmatrix} 1 & 2 \\ 4 & -6 \end{pmatrix}$ in matrix notation.

Similarly, the coefficients of p , q and r in the equation

$$2p - 3q + 4r = 1$$

$$-p + 8q - 5r = 3$$

$$6p + 3q - r = 2$$

The vertical arrangement of a given matrices is called **Column** while the horizontal arrangement is called **Row**

ORDER OF A MATRIX

2 by 2 order of a matrix e. g $\begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}$ can be denoted as $\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$

Where a_{11} is the element in first row, first column

a_{12} is the element in first row , second column

a_{21} is the element in second row , first column

a_{22} is the element in second row , second column

$\Rightarrow$ 3 by 3 order of a matrix e. g.

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \Rightarrow \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$\Rightarrow$ 2 by 3 order of a matrix e.g.

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \Rightarrow \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix}$$

The number of rows in a matrix is usually specified by M and the number of columns by n and a matrix referred to as “ m by n ” matrix.

ADDITION, SUBTRACTION AND MULTIPLICATION OF MATRICES

I. ADDITION OF MATRICES

Corresponding elements in two matrices may be added to form a single matrix.

Example

Add this matrices $\begin{pmatrix} 2 & -1 \\ -7 & 4 \end{pmatrix}$ and $\begin{pmatrix} -3 & 0 \\ 7 & -4 \end{pmatrix}$

Solution

Adding the corresponding elements of the matrices gives

$$\begin{pmatrix} 2 & -1 \\ -7 & 4 \end{pmatrix} + \begin{pmatrix} -3 & 0 \\ 7 & -4 \end{pmatrix} = \begin{pmatrix} 2+(-3) & -1+0 \\ -7+4 & -4+4 \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ 0 & 0 \end{pmatrix}$$

II. SUBTRACTION OF MATRICES

If x is a matrix and y is another matrix, the $(x - y)$ is a single matrix found by subtracting the elements of y from the corresponding elements of x .

Example:

Subtract $\begin{pmatrix} -3 & 0 \\ 7 & -4 \end{pmatrix}$ and $\begin{pmatrix} 4 & -2 \\ 3 & 6 \end{pmatrix}$

Solution

To find matrix $x - y$, the elements of y are taken from the corresponding elements of x . Thus

$$\begin{pmatrix} 4 & -2 \\ 3 & 6 \end{pmatrix} - \begin{pmatrix} -3 & 0 \\ 7 & -4 \end{pmatrix} = \begin{pmatrix} 4 - (-3) & -2 - 0 \\ 3 - 7 & 6 - (-4) \end{pmatrix} = \begin{pmatrix} 7 & -2 \\ -4 & 10 \end{pmatrix}$$

Example

If $A = \begin{pmatrix} -3 & 0 \\ 7 & -4 \end{pmatrix}$, $B = \begin{pmatrix} 2 & -1 \\ -7 & 4 \end{pmatrix}$, and $C = \begin{pmatrix} 1 & 0 \\ -2 & -4 \end{pmatrix}$. Find $A + B - C$

Solution.

$$\begin{pmatrix} -3 & 0 \\ 7 & -4 \end{pmatrix} + \begin{pmatrix} 2 & -1 \\ -7 & 4 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ -2 & -4 \end{pmatrix} = \begin{pmatrix} -3+2-1 & 0+2-0 \\ 7+(-7)-(-2) & -4+4-(-4) \end{pmatrix} = \begin{pmatrix} -2 & -1 \\ 2 & 4 \end{pmatrix}$$

III. MULTIPLICATION OF MATRICES

When a matrix is multiplied by a number, a single matrix results in which each element of the original matrix has been multiplied by the number.

Example 1.

Let $A = \begin{pmatrix} -3 & 2 \\ 6 & 5 \end{pmatrix}$ and $B = \begin{pmatrix} 6 & 4 \\ -1 & 2 \end{pmatrix}$. Find (i), $2A + B$, (ii) $3B - A$

Solution

$$2A = 2 \begin{pmatrix} -3 & 2 \\ 6 & 5 \end{pmatrix} = \begin{pmatrix} -6 & 4 \\ 12 & 10 \end{pmatrix}. \text{ Hence, } 2A + B = \begin{pmatrix} -6 & 4 \\ 12 & 10 \end{pmatrix} + \begin{pmatrix} 6 & 4 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 8 \\ 11 & 12 \end{pmatrix}$$

(ii)

$$3B - A = 3 \begin{pmatrix} 6 & 4 \\ -1 & 2 \end{pmatrix} - \begin{pmatrix} -3 & 2 \\ 6 & 5 \end{pmatrix} = \begin{pmatrix} 18 & 12 \\ -3 & 6 \end{pmatrix} - \begin{pmatrix} -3 & 2 \\ 6 & 5 \end{pmatrix}$$

$$= \begin{pmatrix} 18 - (-3) & 12 - 2 \\ -3 - 6 & 6 - 5 \end{pmatrix} = \begin{pmatrix} 21 & 10 \\ -9 & 1 \end{pmatrix}$$

Example 2.

If $A = \begin{pmatrix} 2 & 3 \\ 1 & -4 \end{pmatrix}$ and $B = \begin{pmatrix} -5 & 7 \\ -3 & 4 \end{pmatrix}$. Find $A \times B$

Solution

Let $A \times B = C$ where $C = \begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix}$

c_{11} is the sum of the products of the first row elements of **A** and the first column elements of **B** taken one at a time.

i.e. $c_{11} = [2 \times (-5)] + [3 \times (-3)] = -19$

c_{12} is the sum of the products of the first row elements of **A** and the second elements of **B** taken one at a time.

i.e. $c_{12} = (2 \times 7) + (3 \times 4) = 26$

c_{21} is the sum of the products of the second row elements of **A** and the first column elements of **B** taken one at a time.

$$c_{21} = [1 \times (-5)] + [-4 \times (-3)] = 7$$

Finally, c_{22} is the sum of the products of the second row elements of **A** and the second column elements of **B**

$$c_{22} = [(1 \times 7) + (-4) \times 4] = -9$$

$$\text{Thus } A \times B = \begin{pmatrix} 2 & 3 \\ 1 & -4 \end{pmatrix} \times \begin{pmatrix} -5 & 7 \\ -3 & 4 \end{pmatrix} = \begin{pmatrix} 19 & 26 \\ 7 & -9 \end{pmatrix}$$

Example 3.

$$\text{If } A = \begin{pmatrix} 3 & 4 & 0 \\ -2 & 6 & -3 \\ 7 & -4 & 1 \end{pmatrix} \text{ and } B = \begin{pmatrix} 2 & -5 \\ 5 & -6 \\ -1 & -7 \end{pmatrix}. \text{ Find } A \times B$$

Solution

NOTE: The sum of the product of the elements of each of the first matrix and the elements of each column of the second matrix are taken at a time.

$$\begin{pmatrix} 3 & 4 & 0 \\ -2 & 6 & -3 \\ 7 & -4 & 1 \end{pmatrix} \times \begin{pmatrix} 2 & -5 \\ 5 & -6 \\ -1 & -7 \end{pmatrix}$$

$$\begin{pmatrix} [(3 \times 2) + (4 \times 5) + (0 \times (-1))] & [(3 \times (-5)) + (4 \times (-6)) + (0 \times (-7))] \\ [((-2) \times 2) + (6 \times 5) + (-3 \times (-1))] & [(-2 \times (-5)) + (6 \times (-6)) + ((-3) \times (-7))] \\ [(7 \times 2) + (-4 \times 5) + (1 \times (-1))] & [(7 \times (-5)) + (-4 \times -6) + (1 \times (-7))] \end{pmatrix}$$

$$\begin{pmatrix} [6 + 20 + 0] & [-15 - 24 + 0] \\ [-4 + 30 + 3] & [10 - 36 + 21] \\ [14 - 20 - 1] & [-35 + 24 - 7] \end{pmatrix}$$

$$\begin{pmatrix} 26 & -39 \\ 29 & -5 \\ -7 & -18 \end{pmatrix}$$

Example 4

$$\begin{pmatrix} 1 & 2 & 1 \\ 1 & 1 & 1 \\ 2 & 1 & 0 \end{pmatrix} \times \begin{pmatrix} 2 & 3 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 2+0+1 & 3+2+1 & 1+0+1 \\ 2+0+1 & 3+1+1 & 1+0+1 \\ 4+0+0 & 6+1+0 & 2+0+0 \end{pmatrix} = \begin{pmatrix} 3 & 6 & 2 \\ 3 & 5 & 2 \\ 4 & 7 & 2 \end{pmatrix}$$

Example 5.

If $A = \begin{pmatrix} 5 & 4 & 2 \\ 3 & 5 & 6 \\ 7 & 2 & 9 \end{pmatrix}$ and $B = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$. Find $A \times B$

Solution.

$$A \times B = \begin{pmatrix} 5 & 4 & 2 \\ 3 & 5 & 6 \\ 7 & 2 & 9 \end{pmatrix} \times \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} (5 \times 1) + (4 \times 2) + (2 \times 1) \\ (3 \times 1) + (5 \times 2) + (6 \times 1) \\ (7 \times 1) + (2 \times 2) + (9 \times 1) \end{pmatrix} = \begin{pmatrix} 5 + 8 + 2 \\ 3 + 10 + 6 \\ 7 + 4 + 9 \end{pmatrix} = \begin{pmatrix} 15 \\ 19 \\ 26 \end{pmatrix}$$

THE UNIT MATRIX

A unit matrix is one in which all elements of the leading diagonal (\) have a value of 1 and all other elements have a value of 0. Multiplication of a matrix by I is equivalent of multiplying by 1 in arithmetic.

DETERMINANT

We have determinant for 2 by 2 and 3 by 3 matrices.

The determinant of 2 by 2 matrix, $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is defined as $(ad - bc)$.

The elements of the determinant of a matrix are written between vertical

lines. Thus, the determinant of $\begin{pmatrix} 3 & -4 \\ 1 & 6 \end{pmatrix}$ is written as $\begin{vmatrix} 3 & -4 \\ 1 & 6 \end{vmatrix}$ and is equal to

$$[(3 \times 6) - (-4 \times 1)] = 18 - (-4) = 22$$

Hence, the determinant of a matrix can be expressed as a single

numerical value. i.e. $\begin{vmatrix} 3 & -4 \\ 1 & 6 \end{vmatrix} = 22$

$\Rightarrow$ The determinant of a 3 by 3 matrix.

Example

Find the determinant of $\begin{vmatrix} 1 & 4 & -3 \\ -5 & 2 & 6 \\ -1 & -4 & 2 \end{vmatrix}$

Solution

Using the first row;

$$\begin{aligned} & 1 \begin{vmatrix} 2 & 6 \\ -4 & 2 \end{vmatrix} - 4 \begin{vmatrix} -5 & 6 \\ -1 & 2 \end{vmatrix} + (-3) \begin{vmatrix} -5 & 2 \\ -4 & -4 \end{vmatrix} \\ &= [(2 \times 2) - (-4 \times 6)] - 4[(-5 \times 2) - (-1 \times 6)] - 3[(-5 \times (-4) - (-1 \times 2))] \\ &= 1(4 + 24) - 4(-10 + 6) - 3(20 + 2) \\ &= 28 + 16 - 66 \\ &= -22 \end{aligned}$$

Hence, the determinant of $\begin{vmatrix} 1 & 4 & -3 \\ -5 & 2 & 6 \\ -1 & -4 & 2 \end{vmatrix} = -22$

CO-FACTORS (2 X 2) ORDER

For each element in any given square matrix, there exist a co-factor, i.e. the minor of the element in determinant together with its place sign.

Example Consider a matrix $A = \begin{vmatrix} 9 & 7 \\ 8 & 6 \end{vmatrix}$

NOTE: The cofactor element A_{ij} is $(-1)^{i+j}$ times the minor of A_{ij} where

the place signs are $\begin{vmatrix} + & - \\ - & + \end{vmatrix}$. The minor of each element is obtained by eliminating the elements in its row and column.

Solution

The minor of $(9) = 6$, $(7) = 8$, $(8) = 7$, $(6) = 9$

$$(6) \times (-1)^{1+1} = (6) \times (-1)^2 = 6$$

$$(8) \times (-1)^{1+2} = (8) \times (-1)^3 = -8$$

$$(7) \times (-1)^{2+1} = (7) \times (-1)^3 = -7$$

$$(9) \times (-1)^{2+2} = (9) \times (-1)^4 = 9$$

$$\text{Hence, cofactor} = \begin{vmatrix} 6 & -8 \\ -7 & 9 \end{vmatrix}$$

However this can also be obtained by “interchange the major diagonal, multiply the minor diagonal by minus sign and find the transpose of the matrix”.

Example 2: Consider a matrix $B = \begin{vmatrix} -3 & 2 \\ 4 & -2 \end{vmatrix}$

Solution

$$(-2) \times (-1)^{1+1} = (-2) \times (-1)^2 = -2$$

$$(4) \times (-1)^{1+2} = (4) \times (-1)^3 = -4$$

$$(2) \times (-1)^{2+1} = (2) \times (-1)^3 = -2$$

$$(-3) \times (-1)^{2+2} = (-3) \times (-1)^4 = -3$$

$$\text{Hence, cofactor} = \begin{vmatrix} -2 & -4 \\ -2 & -3 \end{vmatrix}$$

COFACTOR (3 X 3) ORDER

$$A = \begin{vmatrix} 1 & 4 & -3 \\ -5 & 2 & 6 \\ -1 & -4 & 2 \end{vmatrix} \quad \text{While the place signs are} \quad \begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix}$$

$$\text{Minors obtained from matrix A are} \quad \begin{vmatrix} 28 & -4 & 22 \\ -4 & -1 & 0 \\ 30 & -9 & 22 \end{vmatrix}$$

While the cofactor = minor x place sign

$$\text{Cofactor} = \begin{vmatrix} 28 & 4 & 22 \\ 4 & -1 & 0 \\ 30 & 9 & 22 \end{vmatrix}$$

Example: Find the cofactors of the elements of the first row of the determinant

$$\begin{vmatrix} 2 & 3 & 5 \\ 4 & 1 & 0 \\ 6 & 2 & 7 \end{vmatrix}$$

Solution

The minor of element (2) of the matrix is $\begin{vmatrix} 1 & 0 \\ 2 & 7 \end{vmatrix} = (1 \times 7 - 0 \times 2) = 7$

The minor of element (3) of the matrix is $\begin{vmatrix} 4 & 0 \\ 6 & 7 \end{vmatrix} = (4 \times 7 - 0 \times 6) = 28$

The minor of element (5) of the matrix is $\begin{vmatrix} 4 & 1 \\ 6 & 2 \end{vmatrix} = (4 \times 2 - 6 \times 1) = 2$

Hence the Cofactor of (2) = +7

Cofactor of (3) = -28

Cofactor of (5) = +2

ADJOINT OF A MATRIX

The transpose of a cofactor is the adjoint of the matrix C denoted by C^T .

Example. Consider

$$C = \begin{vmatrix} 1 & 4 & -3 \\ -5 & 2 & 6 \\ -1 & -4 & 2 \end{vmatrix}$$

The cofactor of C is $\begin{vmatrix} 28 & 4 & 22 \\ 4 & -1 & 0 \\ 30 & 9 & 22 \end{vmatrix}$

$$\text{Adjoint of } C = \begin{vmatrix} 28 & 4 & 30 \\ 4 & -1 & 9 \\ 22 & 0 & 22 \end{vmatrix}$$

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INVERSE OF A MATRIX

Find the inverse of the matrix $\begin{vmatrix} 3 & 4 & -1 \\ 2 & 0 & 7 \\ 1 & -3 & -2 \end{vmatrix}$

Solution

The inverse of matrix A, is denoted as $A^{-1} = \frac{\text{adjoint } A}{|A|}$

- i. Obtaining the matrix of the cofactors of the elements
- ii. Transposing this matrix.

The cofactor of element 3 is $\begin{vmatrix} 0 & 7 \\ -3 & -2 \end{vmatrix} = 21$

The cofactor of element 4 is $\begin{vmatrix} 2 & 7 \\ 1 & -2 \end{vmatrix} = 11$ and so on. The matrix of

cofactor is $\begin{vmatrix} 21 & 11 & -6 \\ 11 & -5 & 13 \\ 28 & -23 & -8 \end{vmatrix}$

The transpose of the matrix of cofactors, i.e the adjoint of the matrix is obtained by writing the rows as columns and is

$$\begin{vmatrix} 21 & 11 & 28 \\ 11 & -5 & -23 \\ 6 & 13 & -8 \end{vmatrix} \text{ And the determinant of } \begin{vmatrix} 3 & 4 & -1 \\ 2 & 0 & 7 \\ 1 & -3 & -2 \end{vmatrix} \text{ is } 113, \text{ i.e. } |A|=113$$

Hence, the inverse is

$$A^{-1} = \frac{\text{adjoint } A}{|A|} = \frac{\begin{vmatrix} 21 & 11 & 28 \\ 11 & -5 & -23 \\ 6 & 13 & -8 \end{vmatrix}}{113}$$

$$= \begin{vmatrix} \frac{21}{113} & \frac{11}{113} & \frac{28}{113} \\ \frac{11}{113} & -\frac{5}{113} & -\frac{23}{113} \\ \frac{6}{113} & \frac{13}{113} & -\frac{8}{113} \end{vmatrix}$$

CRAMMERS RULE

Consider a set of simultaneous linear equations

$$ax_1 + bx_2 + cx_3 = y_1$$

$$dx_1 + ex_2 + fx_3 = y_2$$

$$gx_1 + hx_2 + ix_3 = y_3$$

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

Where **x** and **y** denotes the unknown variables and constants of the equation.

Procedure.

1. Find the determinant of the given matrix $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$

2. Interchange the column of the matrix by the constants respectively to obtain a new matrix whose determinants are resolved for similarly. Say

$$B = \begin{bmatrix} y_1 & b & c \\ y_2 & e & f \\ y_3 & h & i \end{bmatrix}, \quad C = \begin{bmatrix} a & y_1 & c \\ d & y_2 & f \\ g & y_3 & i \end{bmatrix}, \quad D = \begin{bmatrix} a & b & y_1 \\ d & e & y_2 \\ g & h & y_3 \end{bmatrix}$$

$$y_1 = \frac{\text{Det} B}{\text{Det} A}, \quad y_2 = \frac{\text{Det} C}{\text{Det} A}, \quad y_3 = \frac{\text{Det} D}{\text{Det} A}$$

Example.

Consider

$$2x_1 + 4x_2 + x_3 = 3$$

$$5x_1 + 6x_2 + 7x_3 = 2$$

$$3x_1 + 2x_2 + x_3 = 1$$

Find the value of the unknown using Cramer's rule.

Solution

$$\begin{vmatrix} 2 & 4 & 1 \\ 5 & 6 & 7 \\ 3 & 2 & 1 \end{vmatrix} \begin{vmatrix} x_1 \\ x_2 \\ x_3 \end{vmatrix} = \begin{vmatrix} 3 \\ 2 \\ 1 \end{vmatrix}$$

$$A = \begin{vmatrix} 2 & 4 & 1 \\ 5 & 6 & 7 \\ 3 & 2 & 1 \end{vmatrix}$$

$$\begin{aligned} |A| &= +2(6 - 14) - 4(5 - 21) + 1(10 - 18) \\ &= 2(-8) - 4(-16) + 1(-8) \\ &= -16 + 64 - 8 \\ &= 40 \end{aligned}$$

To obtain the values of the unknown variables, we have;

$$B = \begin{vmatrix} 3 & 4 & 1 \\ 2 & 6 & 7 \\ 1 & 2 & 1 \end{vmatrix} = 3(-8) - 4(-5) + 1(-12) = -6$$

$$C = \begin{vmatrix} 2 & 3 & 1 \\ 5 & 2 & 7 \\ 3 & 1 & 1 \end{vmatrix} = 2(-5) - 3(-16) + (-1) = 37$$

$$D = \begin{vmatrix} 2 & 4 & 3 \\ 5 & 6 & 2 \\ 3 & 2 & 1 \end{vmatrix} = 2(2) - 4(-1) + 3(-8) = 16$$

$$x_1 = \frac{\text{Det } B}{|A|} = \frac{-6}{40} = \frac{-3}{20}$$

$$x_2 = \frac{\text{Det } C}{|A|} = \frac{37}{40} = \frac{37}{20}$$

$$x_3 = \frac{\text{Det } D}{|A|} = \frac{-16}{40} = \frac{-2}{5}$$

TUTORIAL QUESTIONS

1. Find the determinant if $A = \begin{vmatrix} 4 & -7 & 6 \\ -2 & 4 & 0 \\ 5 & 7 & -4 \end{vmatrix}$

2. Evaluate $\begin{vmatrix} 8 & -2 & -10 \\ 2 & -3 & -2 \\ 6 & 3 & 8 \end{vmatrix}$

3. Find the transpose of $\begin{vmatrix} 4 & -7 & 6 \\ -2 & 4 & 0 \\ 5 & 7 & -4 \end{vmatrix}$

4. Find the cofactor of $\begin{vmatrix} 1 & 2 & 1 \\ 4 & 3 & 2 \\ 1 & 2 & 3 \end{vmatrix}$

5. Find the minor of $\begin{vmatrix} 6 & 4 & 2 \\ 1 & 2 & 3 \\ 4 & 2 & 1 \end{vmatrix}$

6. Determine the adjoint of $\begin{vmatrix} -2 & 6 & -4 \\ 2 & 0 & -8 \\ -1 & 7 & 3 \end{vmatrix}$

THANK YOU FOR LISTENING.